

A Weymouth

mock Exam - Taken 9.23.26 @ 1200

- Real Exam 9.29.26 @ 1130 → 75 min limit.

Derivation 1: 2 particle System w/ Equations of motions

$\vec{R}$ position of COM, $M = m_1 + m_2$ (total mass) $M = m_1 + m_2$.

$$i) M \frac{d^2 \vec{R}}{dt^2} = \sum_i \vec{F}_i^{(e)} = \vec{F}^{(e)}$$

and

$$ii) \frac{d\vec{L}}{dt} = \sum_i \vec{N}_i^{(e)} = \vec{N}^{(e)}$$

→ Show how internal forces obey weak law of interaction (a) and Strong (b)

$t = 0 / 75 \text{ min}:$

↳ Inertial frame, let L or r be about some fixed origin

$\vec{F}_{12}$ is force on particle 1 due to 2, constant mass

↳ Newton:

$$m_1 \ddot{\vec{r}}_1 = \vec{F}_1^{(e)} + \vec{F}_{12}^{(e)}, \quad m_2 \ddot{\vec{r}}_2 = \vec{F}_2^{(e)} + \vec{F}_{21}^{(e)}$$

(a) Since $M\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2$, → add

$$M\ddot{\vec{R}} = \vec{F}^{(e)} + \vec{F}_{12} + \vec{F}_{21} \dots (1)$$

Comparing w/ given $M\ddot{\vec{R}} = \vec{F}^{(e)}$ yields

$$\vec{F}_{21} = -\vec{F}_{12}$$

thus internal total force are equal and opposite (weak law)

(b) Total angular momentum:

$$\vec{L} = \sum_i \vec{r}_i \times m_i \dot{\vec{r}}_i \quad \text{and bc } \vec{r}_i \times m_i \dot{\vec{r}}_i = 0 \text{ by def of COM}$$

$$\begin{aligned} \dot{\vec{L}} &= \vec{N}^{(e)} + \vec{r}_1 \times \vec{F}_{12} + \vec{r}_2 \times \vec{F}_{21} \\ &= \vec{N}^{(e)} + (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} \end{aligned}$$

Using fact that $\dot{\vec{L}} = \vec{N}^{(e)}$ then

$$(\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} = 0$$

for distinct particles: $\vec{F}_{12} = f(\vec{r}_1 - \vec{r}_2)$ for some scalar f

↳ forces are equal and opposite along path joining particles → Strong law.

(22 min elapsed)

2. Derivation 2:

- (a): circular + parabolic orbits $V(r) = -K/r$; Show perihelion of distance of parabola is $1/2 r_{\text{circular}}$
- (b) Prove Speed of particle at any point is $\sqrt{2} \cdot \bar{v}_{\text{circular}}$ passing thru same point
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a. $V(r) = -K/r, K > 0$

$$V(\infty) = V(-\infty) = 0$$

Let μ be orbiting mass / reduced mass for 2 body motion

Let $J = \|\vec{L}\|$

Via Conservation of Energy

$$E = \frac{\mu}{2} \dot{r}^2 + \frac{J^2}{2\mu r^2} - \frac{K}{r}$$

$\hookrightarrow$ w/ circle effective potential is stationary

$$0 = \frac{d}{dr} \left(\frac{J^2}{2\mu r^2} - \frac{K}{r} \right) \Big|_{r_c} = -\frac{J^2}{\mu r_c^3} + \frac{K}{r_c^2}$$

$$\hookrightarrow r_c = \frac{J^2}{\mu K} \quad \dots (1)$$

A parabolic orbit has $E=0$. At perihelion $r_p = \dot{r} = 0$

$\hookrightarrow$ So for same non-zero J ,

$$0 = \frac{J^2}{2\mu r_p^2} - \frac{K}{r_p} \Rightarrow \boxed{r_p = \frac{J^2}{2\mu K} = \frac{r_c}{2}}$$

b. At any radius r on the parabola, Zero Energy gives:

$$\frac{1}{2} \mu v_p^2 - \frac{K}{r} = 0 \rightarrow v_p^2 = \frac{2K}{\mu r}$$

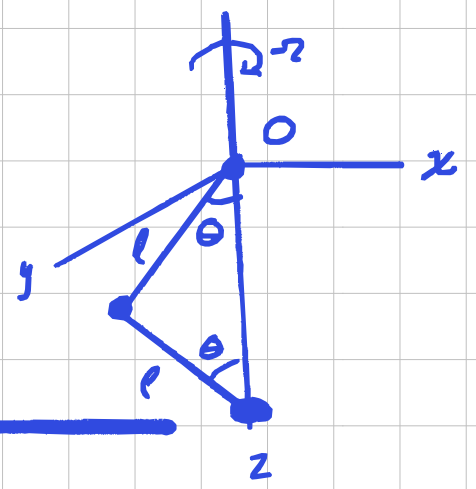
for a circular orbit of same radius, centripetal balance yields:

$$\frac{\mu v_c^2}{r} = \frac{K}{r^2} \rightarrow v_c^2 = \frac{K}{\mu r} \Rightarrow \boxed{v_p = \sqrt{2} \cdot v_c}$$

(43 / 75 min)

Problem 1: rotating two rod system:

- 2 beads, velo Ω about z -axis, constant
- θ describes angle w/ vertical, is
 $\hookrightarrow$ single generalized coord.



a. write Lagrange for θ

- z is positioned down
- then azimuth $\phi = \Omega \cdot t$

beads have positions

$$\vec{r}_1 = (l \sin \theta \cdot \cos(\Omega t), l \sin \theta \sin(\Omega t), l \cos \theta)$$

$$\vec{r}_2 = (0, 0, 2l \cos \theta)$$

$\hookrightarrow$ both rods share length l . then velocities satisfy

$$\vec{v}_1^2 = l^2 \dot{\theta}^2 + l^2 \Omega^2 \sin^2 \theta$$

$$\vec{v}_2^2 = 4l^2 \sin^2 \theta \dot{\theta}^2$$

$$T = \frac{m l^2}{2} [(1 + 4 \sin^2 \theta) \dot{\theta}^2 + \Omega^2 \sin^2 \theta]$$

$$V = -m g (z_1 + z_2) = -3 m g l \cos \theta \quad (+z \text{ is down})$$

$$L = T - V = \frac{m l^2}{2} [(1 + 4 \sin^2 \theta) \dot{\theta}^2 + \Omega^2 \sin^2 \theta] + 3 m g l \cos \theta \quad \dots (1)$$

$$\frac{\partial L}{\partial \dot{\theta}} = m l^2 (1 + 4 \sin^2 \theta) \dot{\theta}$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} &= m l^2 \left[(1 + 4 \sin^2 \theta) \ddot{\theta} + \frac{d}{dt} (1 + 4 \sin^2 \theta) \dot{\theta} \right] \\ &= m l^2 \left[(1 + 4 \sin^2 \theta) \ddot{\theta} + 8 \sin \theta \cos \theta \cdot \dot{\theta}^2 \right] \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial \theta} \left[\frac{m l^2}{2} (1 + 4 \sin^2 \theta) \dot{\theta}^2 \right] &= \frac{m l^2}{2} (8 \sin \theta \cos \theta) \dot{\theta}^2 \\ &= 4 m l^2 \sin \theta \cos \theta \dot{\theta}^2 \end{aligned}$$

$$\text{and } \frac{\partial}{\partial \theta} \left(\frac{m l^2}{2} \Omega^2 \sin^2 \theta \right) = m l^2 \Omega^2 \sin \theta \cos \theta$$

$$\frac{\partial}{\partial \theta} (3 m g l \cos \theta) = -3 m g l \sin \theta$$

$$\hookrightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

$$(1 + 4 \sin^2 \theta) \ddot{\theta} + 4 \sin \theta \cos \theta \dot{\theta}^2 + \left(\frac{3g}{l} - \Omega^2 \cos \theta \sin \theta \right) = 0$$

Pr 1: (b)

$$\text{Equilibrium: } \dot{\theta} = \ddot{\theta} = 0$$

$$\hookrightarrow \text{from pt (a): } \sin \theta \left(\frac{3g}{\ell} - \Omega^2 \cos \theta \right) = 0$$

hence

$$\theta = 0, \quad \theta = \pi \quad \text{and when } \Omega^2 \geq \frac{3g}{\ell}$$

$$\text{there is also } \cos \theta_c = \frac{3g}{\ell \Omega^2}$$

$$\hookrightarrow \theta_c = \cos^{-1} \left(\frac{3g}{\ell \Omega^2} \right)$$

$$\text{Let } \Omega_c = \sqrt{\frac{3g}{\ell}}$$

If:

$$\Omega < \Omega_c : 2 \text{ vertical equilibrium: } \theta = 0, \pi$$

$$\Omega > \Omega_c : \text{also a tilted equilibrium: } \theta_c = \theta$$

In limiting case

$$\Omega \rightarrow 0 : \theta = 0 : \text{Stable config, both rods hang down vertically w/ bead at } z = l$$

$$\text{fast rotation: } \Omega \rightarrow \infty : \cos \theta_c = \frac{3g}{\ell \Omega^2} \rightarrow 0$$

then $\theta_c = \pi/2$ and rod approaches horizontal while bead approaches origin along z -axis.